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Meaning: For each plaintext $x \in \mathcal{P}$, if we encrypt x using a key $k \in \mathcal{K}$ to obtain a ciphertext

$$y := e_k(x),$$

then we may decrypt y to recover

$$x = d_k(y) (= d_k(e_k(x))).$$

Ex: (Shift ciphers as cryptosystems)

- Set of plaintexts $\mathcal{P}$ = set of ciphertexts $\mathcal{C} = \{A, \dots, Z\}$.
- Key space $\mathcal{K} = \{0, \dots, 25\}$ (= genuinely different shifts).
- e_k = cyclic right-shift, k steps
- d_k = cyclic left-shift, k steps

Note that in this case, we're really applying our cryptosystem to strings over the alphabet $\{A, \dots, Z\}$, one character at a time. We'll see later that this is not the only possibility!

Remark: The formal definition of a cryptosystem doesn't take into account "security". For example, here's a useless cryptosystem:

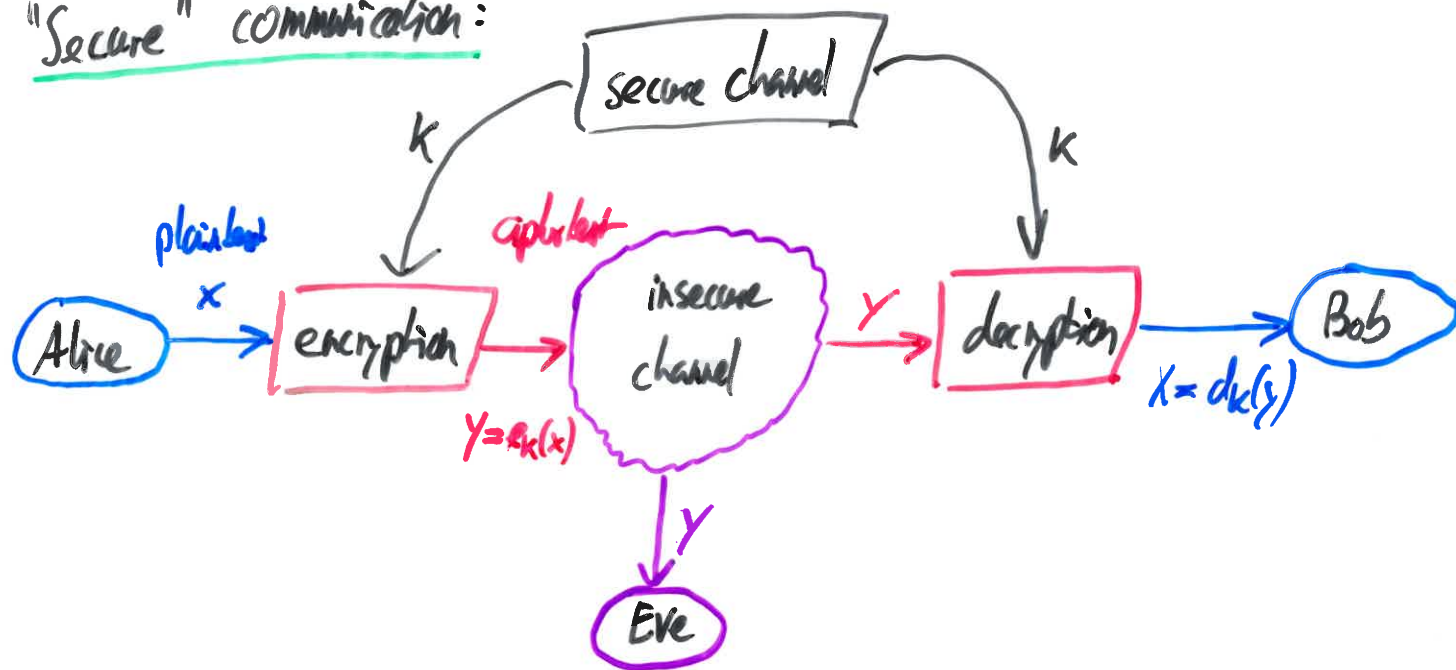
$$\mathcal{P} = \mathcal{C} = \text{any set}, \quad \mathcal{K} = \{0\} \text{ (singleton)}, \quad e_0(x) = x, \quad d_0(y) = y \\ (x, y \in \mathcal{P} = \mathcal{C}).$$

Using cryptography, a schematic

Insecure communication:



"Secure" communication:



Q: Are shift ciphers secure?

Kerckhoff's principle (19th century)

When we study the security of a cryptosystem, we assume that an adversary (e.g. an eavesdropper) knows exactly which cryptosystem is being used but that (s)he does not know the key being used.

In this way, the overall security depends entirely on mathematical properties of the cryptosystem and the plaintexts and keys being used rather than on our ability to keep the encryption and decryption functions secret.

(→ "Security through obscurity"... widely dismissed!)

Q: What does security mean?

We'll assume that we can efficiently compute the encryption and decryption functions corresponding to any given key.

Informally (!), a cryptosystem is secure if it is infeasible for an attacker to decrypt messages without knowing the corresponding key.

Immediate issues

- What does "infeasible" mean? $\rightarrow$ Complexity Theory.
- What does the attacker know? What can (s)he do?
(Intercept some plain texts, or some ciphertexts, or both?
Choose plaintexts and encrypt them?)
- Does the attacker wish to decrypt all types of messages or only some types? (Human-readable text? Random data?)

Now: one shift cipher secure?

Given our alphabet $P = \{A, \dots, Z\}$, there are only 26 possible keys. If our plaintext message is a text (e.g. in English) we can just try out all keys and (probably) guess the original message. $\rightarrow$ "Brute force attack".

This is feasible because our key space is tiny.

Q: Can we increase the size of the key space?

(A sufficiently large key space might render brute force attacks infeasible.)